

Nazmus Sakib

Computer Science & Engineering Student | AI · Full-Stack · RAG Systems

nazmussakib@gmail.com · github.com/nazmussakib · linkedin.com/in/nazmussakib · Dhaka, Bangladesh

EDUCATION

North South University

2021 – 2025

B.Sc. in Computer Science & Engineering | CGPA: 3.10 / 4.00

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, Java, PHP
Frontend:	React 18, Next.js, Vite, Three.js, HTML/CSS
Backend:	FastAPI, Node.js, REST APIs
AI / ML:	Machine Learning, Deep Learning, RAG, YOLOv8, FAISS, BM25, NLP
Databases:	PostgreSQL, MySQL
Tools & Cloud:	Git, Docker, Supabase, JWT Auth, Progressive Web Apps (PWA)

FEATURED PROJECTS

AinSathi (■■■■■■■■) — AI Legal Assistant

Next.js · FastAPI · RAG · FAISS · BM25 · Supabase · Three.js

- Built a production-grade AI legal assistant tailored to Bangladesh's legal framework, supporting bilingual Bengali & English queries.
- Implemented hybrid RAG retrieval pipeline using FAISS vector search + BM25 lexical ranking for accurate, cited legal insights.
- Designed an immersive 3D 'Temple of Justice' UI with Three.js; secured with JWT auth, encrypted sessions, and a privacy mode.

Ratatouille AI — Intelligent Recipe Discovery Platform

React 18 · Vite · TypeScript · FastAPI · YOLOv8 · PWA

- Developed a computer-vision-powered platform that detects ingredients via YOLOv8 object detection and suggests matching recipes in real time.
- Built a smart matching algorithm against a comprehensive recipe database; shipped as an installable, offline-ready Progressive Web App.
- Achieved production-ready standards with optimised performance, comprehensive error handling, and security best practices.

EasyRide — Bus Ticket Management System

PHP · MySQL · OTP · QR Code · Web

- Designed a full-featured web platform automating bus ticket booking, route management, and passenger verification via QR codes.
- Implemented OTP-based confirmation, booking history, and an admin dashboard for route assignment, reporting, and user issue management.
- Integrated chatbot support and real-time tracking to minimise human errors and improve overall passenger experience.

RELEVANT COURSEWORK

Data Structures & Algorithms · Database Systems · Artificial Intelligence · Machine Learning · Computer Networks · Software Engineering · Operating Systems · Computer Vision

LANGUAGES

Bengali (Native) · English (Professional Proficiency)